



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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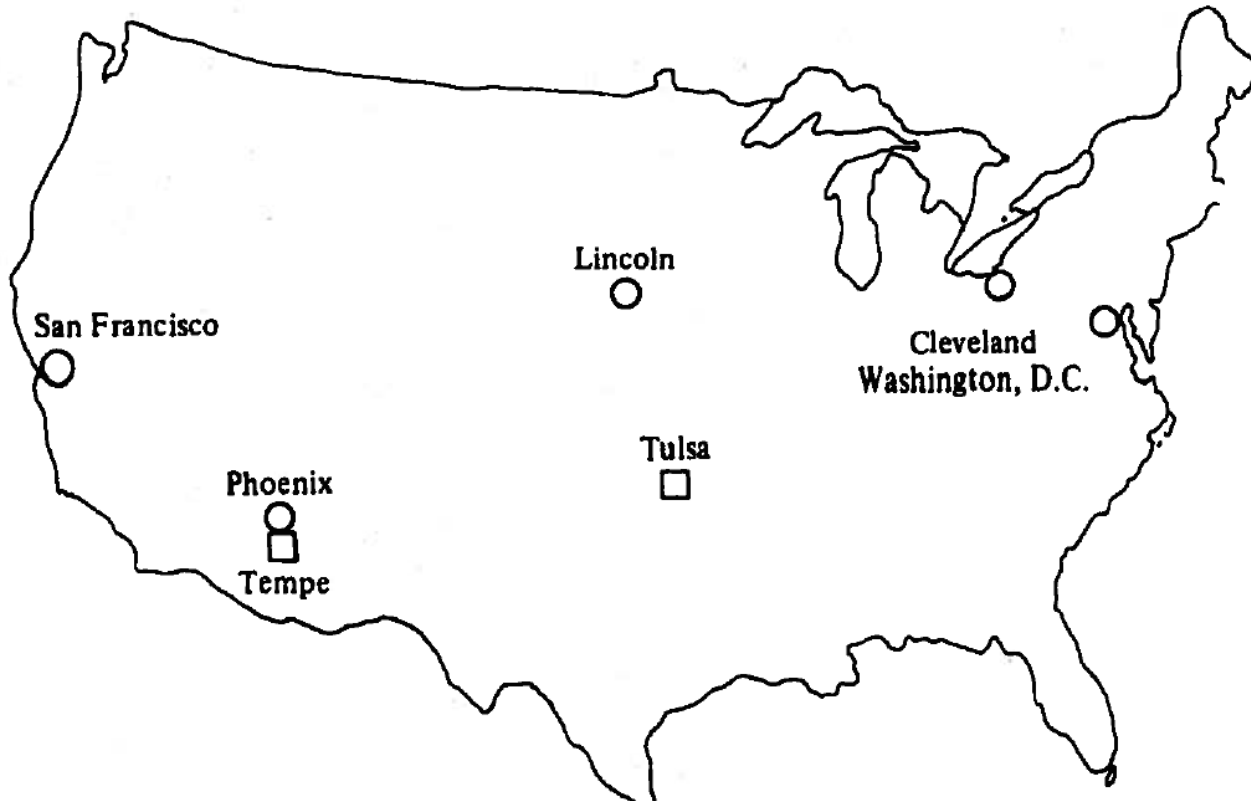


Agenda

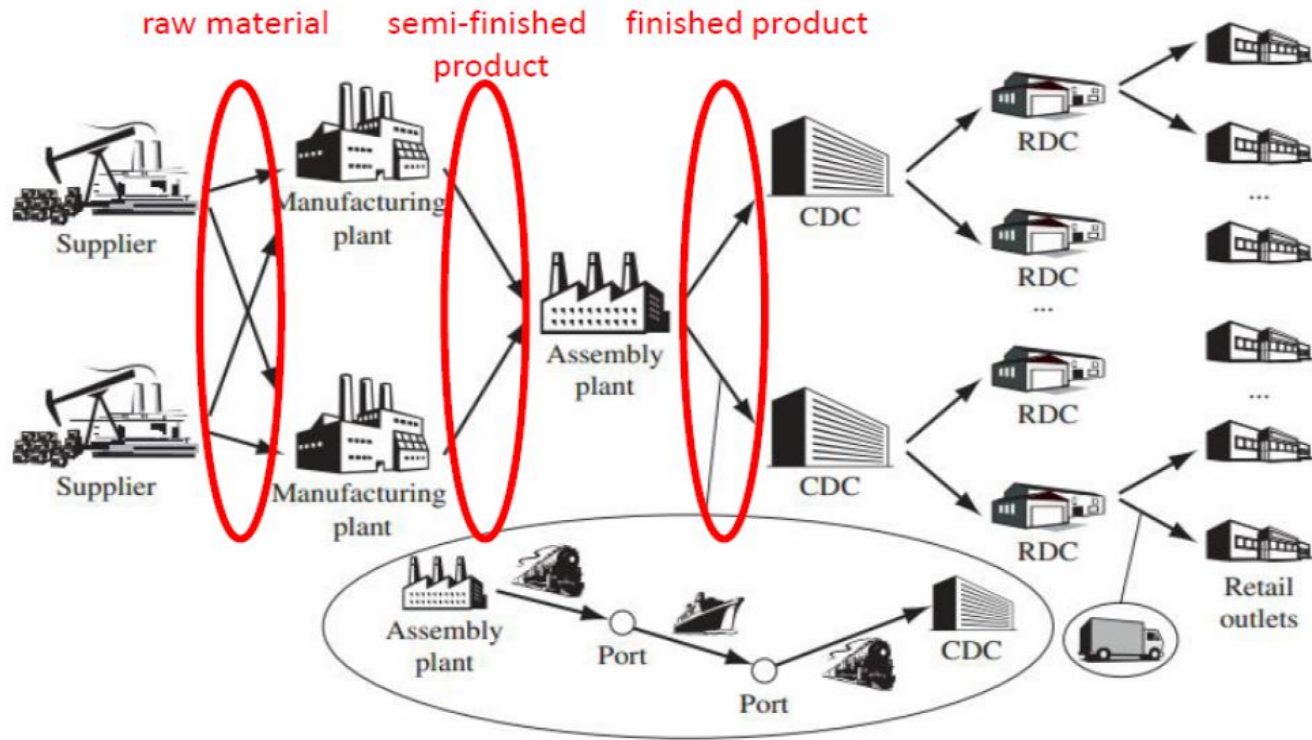
- Chapter 4 - Facilities Location

Facility Location - example

- New plant because of increase in the demand over the next 10 years
- Products must be transported to warehouses within a week



Example – Supply Chain Design



CDC: Central Distribution Centre
RDC: Regional Distribution Centre

Facility Location

- General location
 - Exact site
 - Long-term and strategic decisions
 - Investment and transportation costs
 - Optimization models
- Internal Location
 - Facility layout
 - Material handling
 - Templates, simulation, optimization
- Team work
 - Accountants, lawyers, marketing experts, executives, consultants, Industrial Engineer

Decision Criteria for Location

- Location of markets (potato chips)
- Raw materials (steel mill)
- Transportation
- Power
- Climate and fuel
- Labor and wages
- Laws and taxation
- Community services and attitudes
- Water and waste
- Other

Classification of facility location problems

- Objective:
 - Minimize cost (distance) function
 - Minimize the maximum distance travelled (public facilities)
- Distance measures
 - Straight-line or Euclidian distance
 - Intercity travel
 - Rectilinear distance
 - Urban setting (streets on the grid)
 - Factory floor (corridors, walk paths)
- Number of facilities
 - Single
 - Multiple

Analytical Tools For Locating Facilities

- Transportation method
- Center of gravity method
- Linear programming (mixed-integer programming)

Transportation Problem

- Applicable to single-location facility location with a cost minimization objective (single –objective)
- Decision variables: number of units to be shipped from each source to each destination
- Goal: minimize shipment costs
- The method can be tried for each candidate location
- The one with the minimum transportation cost will be selected

Transportation Model Example

The Plain View Manufacturing Company presently has two factories at Amarillo and Waco, Texas, and three warehouses at San Antonio, Dallas, and Houston, Texas. In recent months, Plain View has been unable to produce and ship enough of its computer peripheral units to satisfy the market demand at the warehouses. A new factory at Huntsville, Texas, is proposed to increase factory capacity. Bill Mayer, Plain View's president, wants to determine what Plain View's monthly shipping costs will be with the new factory located at Huntsville.

Factory	Monthly Capacity (units)	Warehouse	Monthly Warehouse Requirements (units)
Amarillo	400	San Antonio	300
Waco	1,000	Dallas	900
Huntsville	600	Houston	800
Total	2,000	Total	2,000

	Transportation Costs		
	Warehouse		
Factory	San Antonio	Dallas	Houston
Amarillo	\$31	\$21	\$42
Waco	20	21	30
Hunstville	23	20	15

Transportation Model Example

- Least-cost assignment method to find a good feasible solution

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31	21	42	400
Waco	20	21	30	1000
Huntsville	23	20	15	600
Warehouse demand	300	900	800	2000

Solution method

- This problem can be solved following one of the rules given below to:
 - Fill up warehouse demands and use up factories supplies one-by-one
 - Select the smallest unit cost to allocate the largest amount to meet the demand or to use up all supply. If there is a tie, pick anyone.

Example solution

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31 400	21 400	42 400	400
Waco	20 300	21 300	30 300	(1000) 700
Huntsville	23 600	20 600	15 600	600
Warehouse demand	300 500	(900) 500	(800) 200	2000

Transportation example result

Total cost:

$$6 \times 15 + 3 \times 20 + 4 \times 21 + 5 \times 21 + 2 \times 30 = \$399(\times 100)$$

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31 400	21 400	42 200	400
Waco	20 300	21 500	30 200	1000
Huntsville	23 600	20 600	15 600	600
Warehouse demand	300	900	800	2000

Transportation example alternative result (following the same rule)

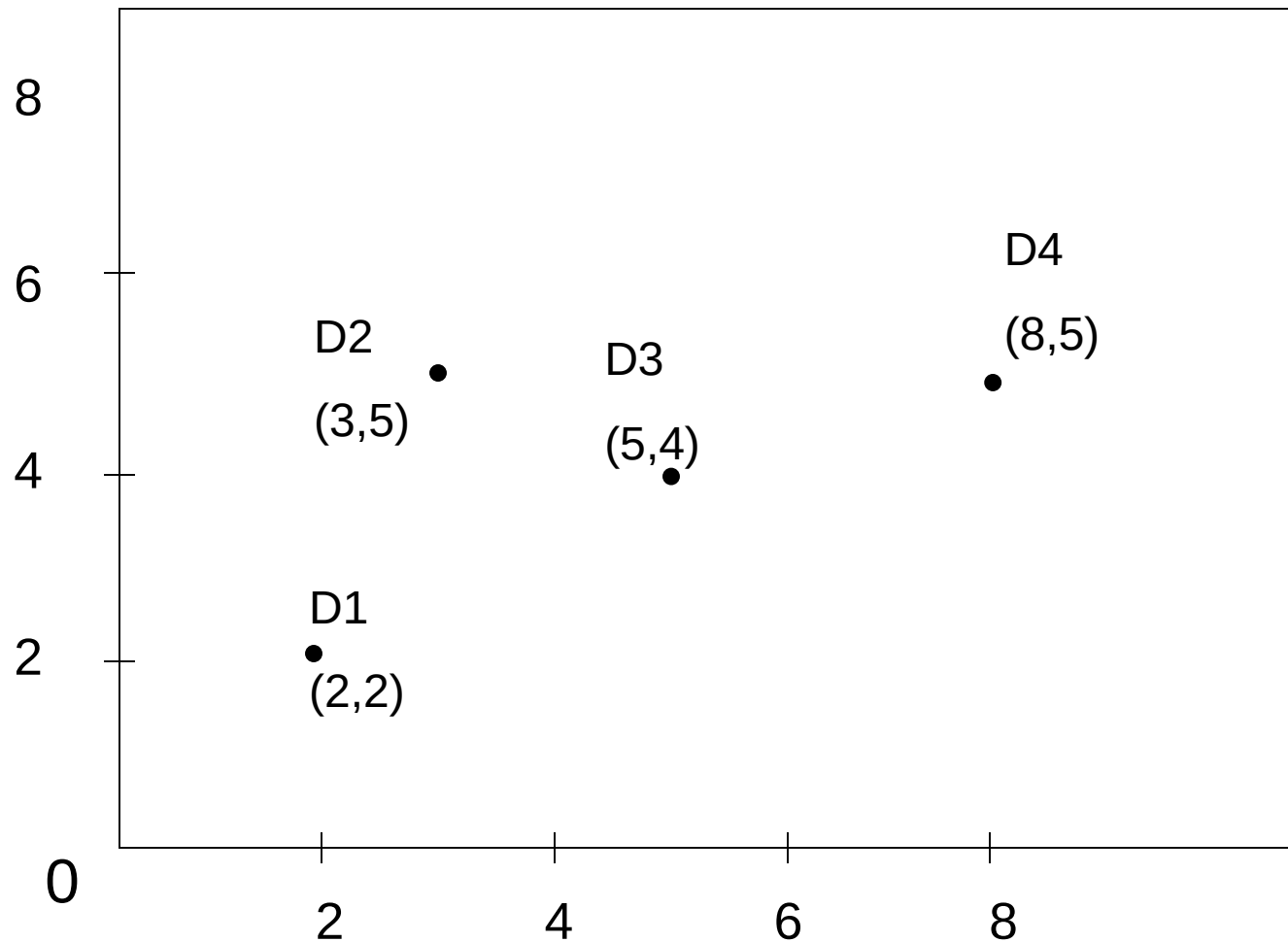
Total cost: $6 \times 15 + 3 \times 20 + 2 \times 21 + 7 \times 21 + 2 \times 42 = \$423 (\times 100)$

From \ To	San Antonio	Dallas	Houston	Factory capacity
Amarillo	31 200	21 200	42 200	400
Waco	300 20 700	21 700	30 600	1000
Huntsville	23 600	20 600	15 600	600
Warehouse demand	300	900	800	2000

Center-of-gravity method

- Location of a blood bank in a city that minimizes distribution costs of shipping blood products to hospitals
- Treats distribution cost as a function of the distance and quantity shipped
- Urban setting → rectilinear distances

Center of Gravity Method



Calculations (equal shipment quantities)

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

- x_i = *x-coordinate of destination i*
- y_i = *y-coordinate of destination i*
- n = *number of destinations (hospitals)*

Calculations (quantities different)

$$\bar{x} = \frac{\sum_{i=1}^n Q_i x_i}{\sum_{i=1}^n Q_i}$$

$$\bar{y} = \frac{\sum_{i=1}^n Q_i y_i}{\sum_{i=1}^n Q_i}$$

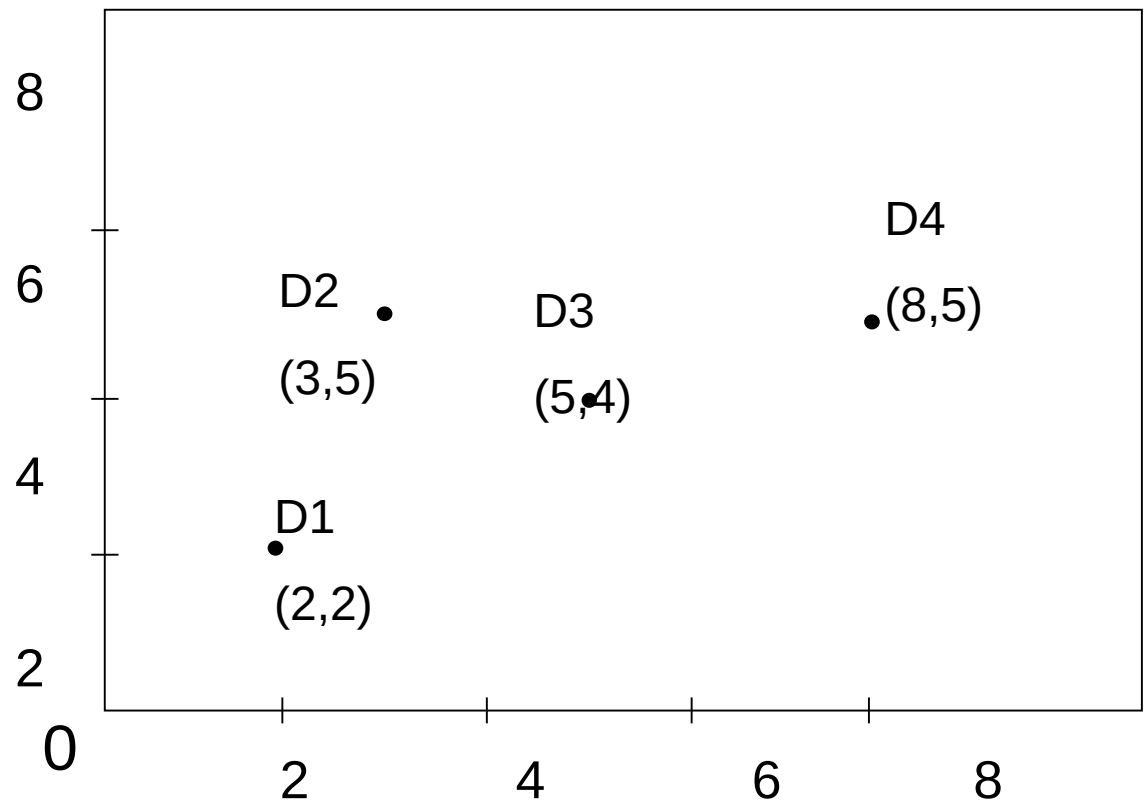
- Q_i = Quantity to be shipped to destination i
- x_i = x -coordinate of destination i
- y_i = y -coordinate of destination i

Alternative method (heuristic)

- Example 4.3
 - Pages 96-99 of text book

Example

- Given the locations of the four destinations shown earlier, find the location of the distribution if the quantities to be shipped are equal, and if the quantities to be shipped vary as follows: D1: 800 units/week, D2: 900 units/week; D3: 200 units/week, D4: 100



Other location problems

- Hub location problems
- Min-max problems
- Emergency service facility location problems
- Quadratic assignment problems
- Balancing between one-time investment costs and transportation (service) costs